Attention Is All You Need

Personal Reflection

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Through our research on the paper *"Attention Is All You Need,"* I learned far more than just one technical architecture.

Even before I started reading and exploring the paper, I had always been interested in the language models we use in our everyday lives. However, I didn't really understand what powered these models or how the history of their development had led to what we have today. Working on this research opened my eyes and sparked my curiosity far beyond what I had imagined.

At first, I had only a vague understanding of the term *Transformer*, mainly as something behind modern language models such as ChatGPT and Gemini. But reading the paper and working on our presentation helped me understand much more deeply why this was such a revolution and what actually happens behind the scenes.

It wasn't just another improvement to existing models. It was a fundamental shift in how sequences of information could be processed, changing the rules of the game.

One of the most important things I learned was why a new model was needed in the first place. I explored the architectures that came before the Transformer, including RNNs, LSTMs, GRUs, and CNNs, and gained a better understanding of their limitations.

Through the paper, I learned how these architectures struggled with parallelization, long-range dependencies, and computational efficiency. The problem wasn't simply that the existing models weren't good enough. Their very structure limited their ability to handle long-distance relationships efficiently.

Another particularly meaningful part of this experience was understanding the Attention mechanism itself.

I learned that the idea wasn't to process words strictly one after another, but to allow each word to look at the other words in a sentence and calculate how relevant they were to it.

What am I looking for? Who might be relevant to me? What information should be passed forward?

Suddenly, the concepts of Query, Key, and Value felt much more meaningful and essential.

I don't just believe that understanding this paper will help me in my future studies. I feel that it already has.

During the same semester, I took a Machine Learning course, and thanks to what I had learned from this paper, I was able to see certain topics from a deeper perspective. Suddenly, different ideas started connecting. It was one of those moments when things just *clicked*, allowing me to recognize connections between concepts that I hadn't previously seen so clearly.

This research brought together several fields that I had studied, or would study, separately: vectors and matrices from linear algebra, probability, computational complexity, machine learning, and many more.

From a professional perspective, I also believe this experience has given me an important foundation.

Today, nearly every area of software development is influenced in some way by language models, recommendation systems, intelligent search, information summarization, and generative AI.

Understanding the ideas behind this paper, even at a conceptual level, allows me to look beyond AI as a *black box* and begin understanding what happens beneath the surface.

I believe this will help me read future research papers, understand new models, and approach AI tools with a more critical perspective.

As someone continuing directly toward a Master's degree in Computer Science, with an interest in conducting research and potentially writing a thesis, this experience has encouraged me to explore the field more deeply.

It has also made me consider pursuing research related to these ideas during my graduate studies, and perhaps even contributing to the development of intelligent systems myself.

Ultimately, the most important lesson I took from this paper is that true innovation sometimes begins with a simple but profound idea:

Don't just learn how things work. Ask why they are built that way, and what could be done differently.